Idea Abstract: YONO-Guard

Theme: Proactive Phishing Mitigation & Zero-Trust APK Integrity Framework

1. Problem Understanding

The primary threat to SBI's digital ecosystem is Social Engineering and Brand Impersonation. Fraudsters exploit psychological triggers such as artificial urgency regarding KYC expiry to bypass security protocols. This manipulates users into side-loading malicious APKs from unofficial sources like WhatsApp. These "look-alike" apps mimic the YONO UI to capture sensitive credentials (OTPs, MPINS, Debit Card details), leading to total account takeover, particularly among users with low digital awareness.

2. The Core Idea: The "Shield-Sync" Framework

We propose YONO-Guard, a multi-layered Zero-Trust security framework designed to shift from reactive blocking to a Proactive Detection & Nudge Engine:

- **Autonomous SMS/URL Intelligence:** An NLP-driven scanner evaluating incoming SMS metadata for urgency triggers and domain mismatches.
- **Cryptographic APK Attestation:** A device-level control performing SHA-256 Hash Verification against the official "Golden Hash" before allowing installation.
- **Vernacular Nudge Architecture:** Real-time, multi-lingual Explainable AI alerts and gamified modules that intervene during high-risk interactions.

3. Innovation & Technical Distinction

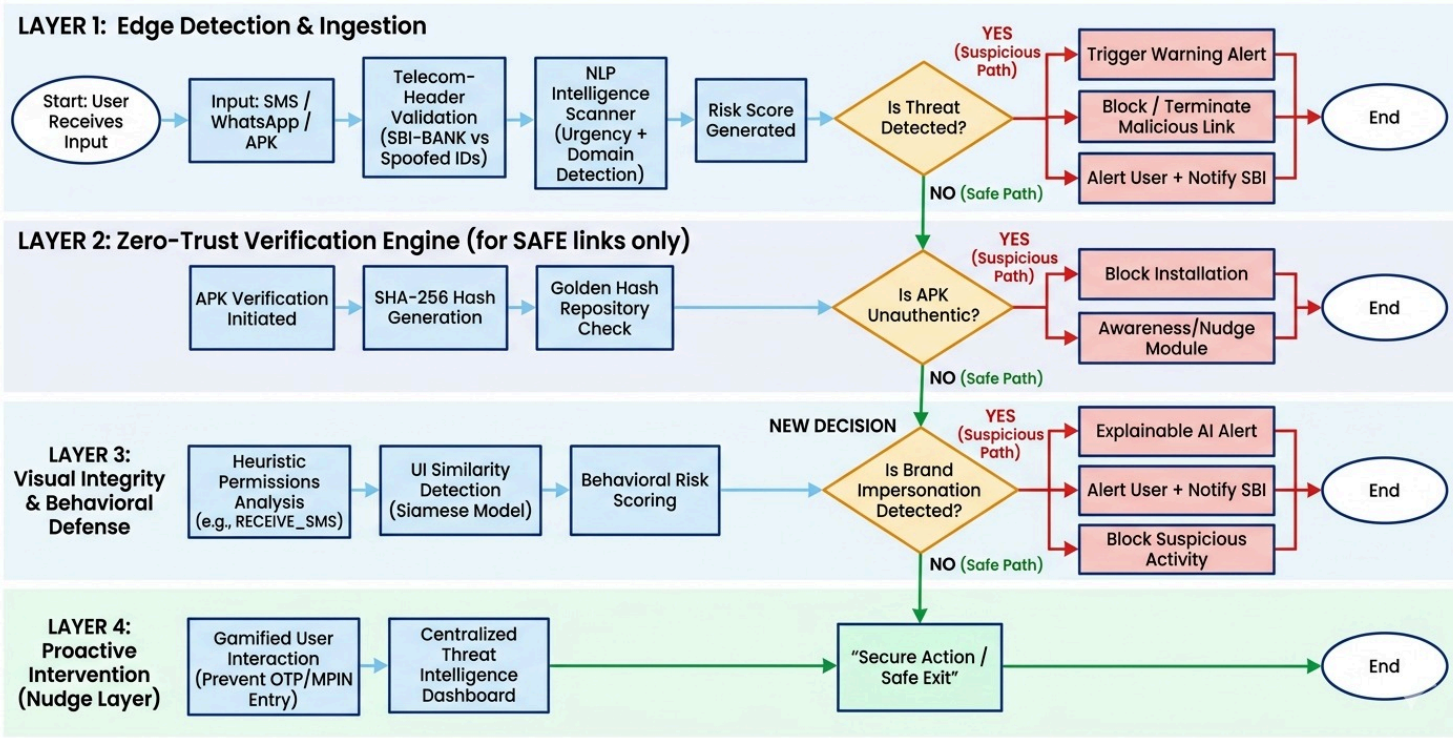
Unlike standard antivirus tools, YONO-Guard targets brand-impersonation logic:

- **Heuristic Permissions Analysis:** Automatically flags apps requesting high-risk permissions (e.g., RECEIVE_SMS) while mimicking official branding.
- **Telecom-Header Validation:** Integration to distinguish official SBI-BANK headers from spoofed alphanumeric IDs.
- **Zero-Trust UI Verification:** Lightweight image-similarity algorithms to detect background apps attempting to overlay or capture the genuine YONO login screen.

4. Desired Impact

Our solution aims for a quantifiable reduction in phishing-originated financial loss, faster removal of fraudulent applications via a centralized intelligence dashboard, and restored digital trust among vulnerable user segments.

5. Team Roles

- **Kingshuk Pattadar & Anishka Agarwal (AI/ML):** Developing Transformer-based NLP and Siamese Networks for phishing and UI-similarity detection.
- **Aayush & Bhuvan Girish Desai (Backend):** Engineering the high-concurrency FastAPI backend and secure "Golden Hash" repository.
- **Afiya Jabeen Kamal (Frontend):** Creating the frictionless vernacular "Nudge" engine and real-time alert interface.

"YONO-Guard: Shield-Sync Architecture Flowchart"